

Feasibility Optimization and Simulations of Hybrid Renewable Energy for Household Electrification in Nigeria

Caleb Akezi Amlabu; Yekini Suberu Mohammed; Abdullahi Shaibu Abubakar;
and Muhammed Jamiu Abdulsalam

Department of Electrical/Electronic Engineering Technology, Federal Polytechnic Nasarawa, Nasarawa State, Nigeria.

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Abstract

This paper executes the design and analysis of a hybrid integrated energy system for the electrification of a rural community. The proposed hybrid system configuration was studied in four different folds using a combination of solar photovoltaic (PV) system, wind turbine (WT), diesel generator (DG), battery (BAT) system, and power converter (CONV) system. Based on the solar radiation, wind energy resources and diesel fuel availability, the optimal sizing configuration of a selected case study rural area was conducted to determine the technoeconomic viability of the rural electrification scheme in a village in northcentral Nigeria through the application of HOMER Pro software. Four different scenarios of PV/BAT/CONV, PV/DG/BAT/CONV, PV/WT/BAT/CONV and PV/WT/DG/BAT/CONV were analyzed for the feasibility study for the case study of Tudu Uku community in Nigeria. The results obtained indicated that the most techno-economically feasible hybrid integrated microgrid design is based on the combination of PV/BAT/CONV with the Net Present Cost (NPC) and the Cost of Energy (COE) of \$11,502 and \$0.216/kWh, respectively.

Keywords: Electricity, Energy, Optimization, Simulation, Rural Electrification, Nigeria.

Introduction

The depletion of fossil fuels and the thoughtful pollution of the global atmosphere, which has a direct impact on climate change, are currently emerging realistic global concerns. Our natural, environmentally friendly landscape is rapidly changing due to climate change restrictions and the resulting global warming scenarios. Global energy stakeholders have been compelled to focus on the utilization of modern Renewable Energy Technologies (RETs) for sustainable development (Pradhan et al., 2017). Apart from the fact that the usage of RETs is becoming more and more popular, there is still another issue that arises from techno-economic difficulties. Over the years, significant figures obtained from the energy sector have promoted the use of RETs to generate electricity. Energy researchers have also stepped up their efforts to investigate the potential switch from traditional energy sources to Renewable Energy Sources (RES). In this context, the application of Solar photovoltaic (PV) and wind energy (WE) applications are typically of great interest in the global energy sector. The electrification of isolated off-grid locations benefits more from the use of RETs for electricity generation.

Due to the high cost of migration and the associated challenging environmental topographies, the majority of rural off-grid localities are deemed economically unworkable for grid extension (Riyatsyah et al., 2022). In certain instances, the potential capability of the electric power produced by solar PV is obstructed by the incidence of intermittent cloud cover and partial shadowing brought on by barriers from buildings and other environmental structures. In addition, the poor performance efficiency of solar PV is a major problem affecting power generation from the system. Furthermore, power generation from the aerodynamics of the atmospheric wind is also another source of renewable electricity, but even if there is natural wind during the day and at night, there are erratic variations in wind speed, which makes its availability questionable. Therefore, to generate stable electricity from solar PV and wind turbines, there must be a mechanism to mitigate the anticipated fluctuations in wind flow and sunlight availability orchestrated from weather fluctuations. In this context, the use of hybrid solar PV or wind turbines, and a diesel generator is frequently used to support supply restrictions. In recent times, diesel generators have been replaced with biomass power generators due to the pronounced environmental pollution associated with diesel generators.

In some locations, biomass is readily available and accessible, and biomass power systems have been used to generate electricity as a replacement for diesel fuel power plants, especially for small-scale electricity. The use of diesel generators in isolated off-grid communities is usually associated with the high cost of energy and complements the dependability constraints imposed by intermittent solar PV and wind turbine energy. The Cost of Energy (CoE) can be influenced by several factors, which must be understood for efficient techno-economic design and analysis of microgrid power plants for rural energy supply. A strategic response to the techno-economic limitations of individual energy sources is the idea of hybridizing several renewable energy components. Figure 1 illustrates how different energy sources can be used to generate power in a traditional microgrid architecture (Ur Rashid et al., 2022). A group of distributed energy resources with integrated Battery Energy Storage Systems (BESS) of various types can be combined to generate electricity for the linked load in the distribution network through the use of microgrids (Abdin and Mérida, 2019). Thus, this study is conducted based on the feasibility assessment and simulation of a Hybrid Renewable Energy System (HRES) for household electrification in Nigeria. The integration of a few renewable energy systems in the form of a microgrid could bring some techno-economic challenges that must be resolved to attain a realistic CoE for the consumers. Therefore, this study will investigate the techno-economic performances of some proposed hybrid microgrid design configurations for the electrification of a small-scale rural community in Nigeria.

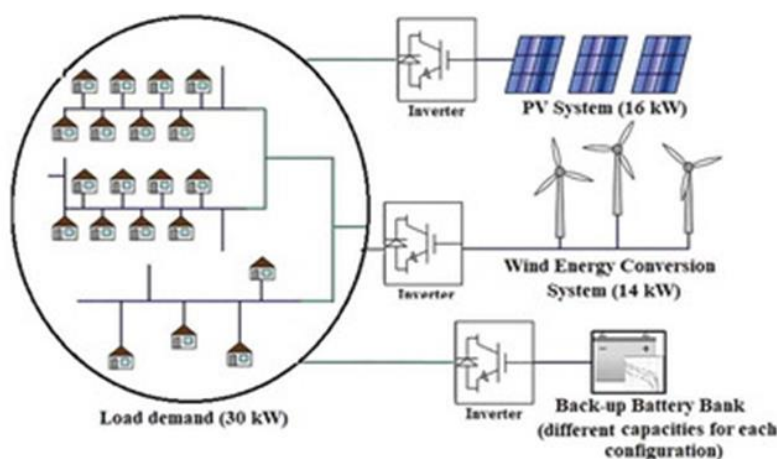


Figure 1: Microgrid architecture

Objectives of the Study

This study aims to investigate the techno-economic reliability of the best hybrid renewable energy configurations for small-scale electrification. The research's primary objectives are:

- To evaluate the rural community's estimated electricity demand.
- To assess the performance and accessibility of renewable energy sources for electricity production in the selected community.
- To investigate the techno-economic aspects of the proposed hybrid configurations and conduct simulations based on the feasibility analysis.

Literature Review

Hybrid Renewable Energy System (HRES) configurations were studied in four Saudi Arabian locations and assessed for feasibility using HOMER software (Tazay, 2020). The study aimed to find economical and environmentally friendly energy options for the project. The simulation's outcome demonstrates that grid-connected photovoltaic (PV) systems are the most common system of power supply for carbon emission reduction and integration of renewable electricity into the national grid in the country. According to Gbadamosi et al. (2022), the most effective way to meet load demand and achieve considerable renewable energy integration is to use an HRES model utilizing renewable energy such as solar PV systems. This approach leads to a significant 32.09% decrease in carbon emissions and a 45% reduction in energy costs. In addition to offering environmental sustainability, this combination makes it financially viable for small-scale electrification projects. In off-grid electric power locations, renewable energy is swiftly taking the lead as the primary substitute for grid-connected electricity. According to this view, the global energy sector is at this time paying close attention to the replacement of fossil fuels with renewable energy sources. This is the justification behind the contemporary global emphasis on renewable energy applications for a reduction in the global energy crisis scenario.

Renewable energy applications have had the opportunity to thrive because of their several benefits, which include improved energy independence, inexhaustibility, and use for rural electrification in far-flung rural localities. The use of hybrid renewable energy systems for electrification has been the subject of numerous contemporary energy studies (Kumar and Sethuraman, 2021; Riayatsyah et al., 2022). The decision-making process about the suitability of hybrid renewable energy systems for rural electrification in Nigeria has been investigated (Faisal and Anwer, 2024; Akram et al., 2020; Murugaperumal and Raj, 2019). Energy is connected to the human standard of living and to reduce poverty in rural communities, and the use of techno-economic sustainability in HRES for electrification must be investigated. Electricity generation using HRES may be techno-economically sustainable, but it depends on a few key parameters, according to previous studies (Halmous et al., 2024; Khalil et al., 2021; Kumar and Channi, 2022). One important area of research is the goal of stable electrification of rural areas with a balance between supply and demand. The sustainability of rural electricity in Nigeria using various hybrid microgrids will thus be examined in this present study. The technical and economic viability of some proposed hybrid microgrids will be studied to provide more insight and informed suggestions into the viability of the project in the country.

Materials and Research Method

HOMER Software Application

Hybrid Optimization Model for Electric Renewables (HOMER) is a renewable energy software developed in the United States by the National Renewable Energy Laboratory (NREL) for standalone and hybrid power plant configurations for techno-economic analysis. The application of this software allows for the design, optimization and simulations of the microgrid system based on renewable energy technologies. It helps to maximize the benefits of a hybrid power system by utilizing standalone microgrids, utility-scale, and distributed generation electric power systems. With energy simulation tools, services, and optimization, HOMER software offers a solid platform for energy researchers to hasten the adoption of distributed and renewable energy sources.

Energy Demand Profile of the Case Study Area

The load profile system of the energy consumption by the subject area is presented in Fig. 2. The figure depicts the average daily load profile in which 16:00 to 21:00 h form the peak demand, which regulates the system size with the electricity demand of 11.27kWh/d. In Fig. 3, the structure of the hybrid power system configuration is shown with the varieties of the power system combination. The seasonal load profile of the case study area is shown in Fig. 4, while Fig. 5 and Fig. 6 show the average monthly solar radiation and the clearness index and the monthly wind speed of the case study location in HOMER Pro. The geographical coordinate of the case study location is 8°32.1'N and 7°28.2'E, Tudu Uku, Nasarawa State

in Nigeria. The solar data resources exhibit the amount of solar radiation with potential to strike the earth surface location of the study area and the data obtained is fed into the HOMER pro software for the generation of the diagram based on the solar radiation intensity as per KWh/m²/day as presented in Fig. 5. Similarly, the Fig. 6 shows the potential wind energy resource concerning the average monthly wind speed of area as plotted in HOMER Pro in m/s covering the period from January to December. The selected site is an off-grid community with low power consumption capacity. The community is well-known for its subsistence agricultural activities like other rural communities close to its location.

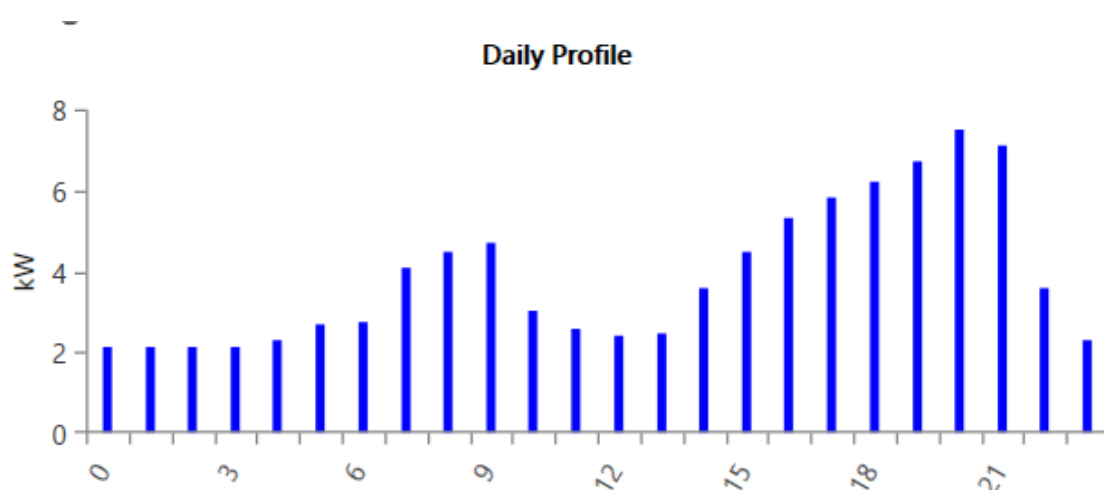


Fig. 2: Load profile of the case study area

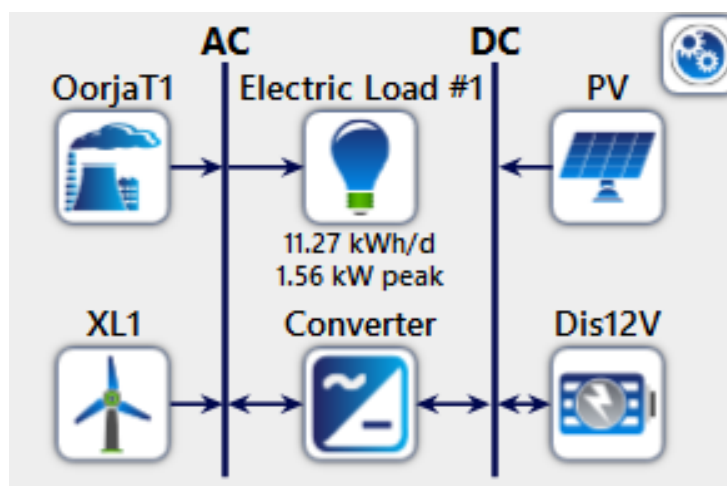


Fig. 3: Structural configuration of the proposed hybrid power system configuration

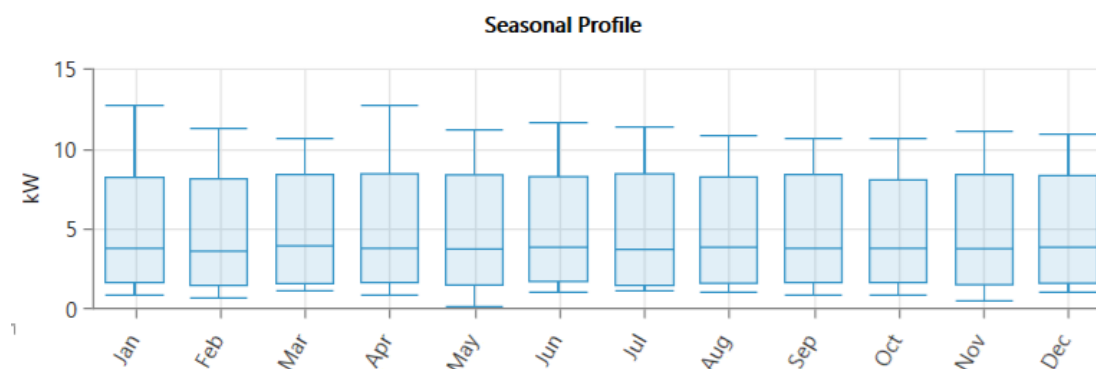


Fig. 4: Seasonal profile of the power demand

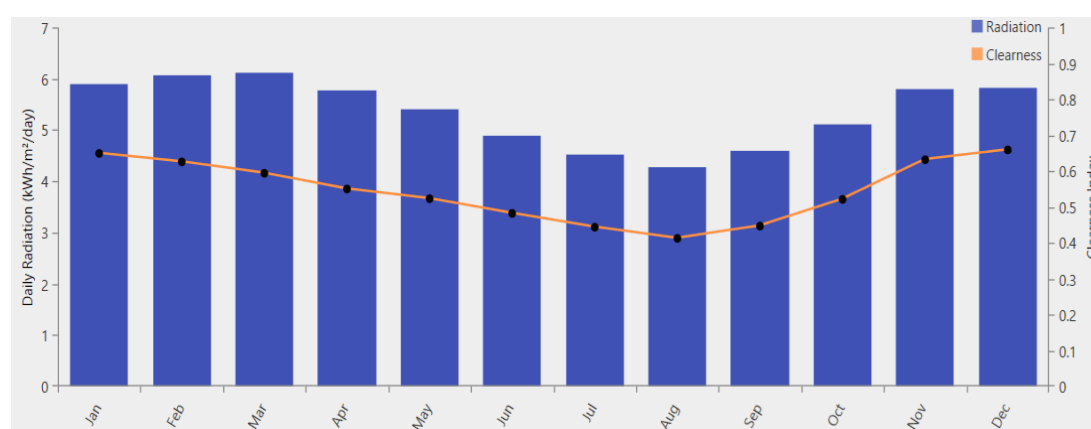


Fig. 5: Average monthly solar radiation and the clearness index

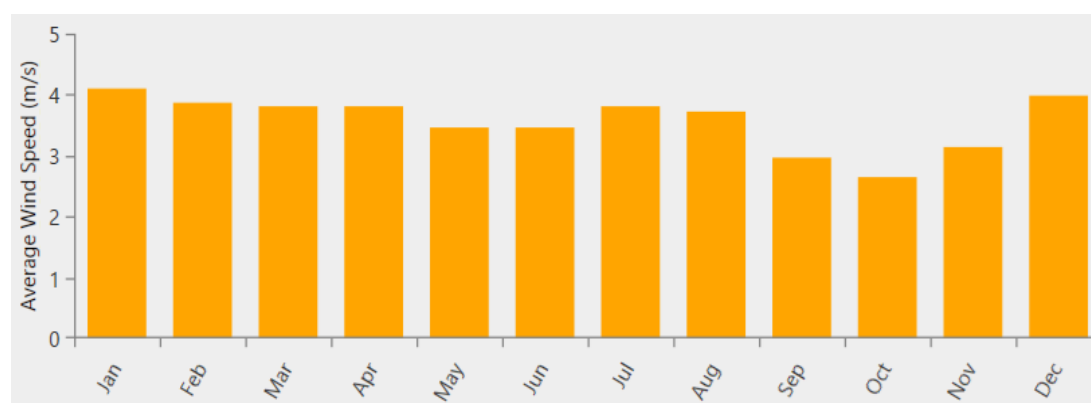


Fig. 6: Average monthly wind speed

Modelling of Solar Energy System

The solar energy system can be modelled for the calculation of the PV output power through the mathematical expression given in Eq. (1). The mathematical expression allows for the computation of maximum output power. Where the abbreviations can be defined as

P_{PV} = maximum power point of a solar PV system, P_M = maximum power point at reference solar radiation, G_r = solar radiation received on module plane, G_R = solar radiation at reference point, τ = temperature correction at maximum power, T_c = module working temperature and T_o = reference temperature (25 °C).

$$P_{PV} = \frac{G_r}{G_R} P_M [1 + \tau(T_c - T_o)] \quad (1)$$

Modeling of Wind Energy System

The mathematical modeling of the electric power production by a wind turbine as shown in Eq. (2):

$$P_{WT} = N_{WT} \begin{cases} 0 & v \leq v_{in} \text{ or } v \geq v_{co} \\ v^3 \left(\frac{P_{WT(rated)}}{v_r^3 - v_{in}^3} \right) - P_{WT(rated)} \left(\frac{v_{in}^3}{v_r^3 - v_{in}^3} \right) & v_{in} \leq v(t) < v_r \\ P_{WT(rated)} & v_r \leq v \leq v_{co} \end{cases} \quad (2)$$

The evaluation of the wind speed (v) can be obtained by using Eq. (3) (Mohammed et al., 2022):

$$v = v_{ref} \times \left(\frac{h_{hub}}{h_{ref}} \right)^\lambda \quad (3)$$

The value of λ is in the range of 0.1 and 0.25 (Mohammed et al., 2022). Where v_r = rated speed, v_{in} = cut-in speed at temperature of 25°C, v_{co} = cut-out speed, v = wind speed, P_{WT} = Power generated by the wind turbine, $P_{WT(rated)}$ = Rated power of the wind turbine, h_{ref} = reference height, h_{hub} = hub height, v_{ref} = velocity at reference height and λ = coefficient of friction of landscape.

Modeling of Diesel Power System

The mathematical expression of the modelling of the diesel power system in HOMER software applies Eq. (4) to obtain the generator's fuel consumption for electric power generation in the framework based on the intercept coefficient of the fuel curve (Mohammed et al., 2022).

$$F = F_o Y_{Gen} + F_1 P_{Gen} \quad (4)$$

Where F_o = fuel curve intercept, Y_{Gen} = rated capacity of the generator (kW), F_1 = fuel slope curve and P_{Gen} = electric power output.

Results and Analysis

Several analyses based on the techno-economic optimization of the different configurations of hybrid microgrids have been studied in the HOMER Pro simulation environment. The optimization of the hybrid microgrid systems presented appears in different categories based on their ranking of techno-economic feasibility, from the most feasible configuration to the least techno-economic system, as shown in Table 1. The NPC and the respective COE of each hybrid system configuration are shown in Table 1, while Table 2 shows the optimized system component sizing and their cost implication.

Table 1: Technoeconomic optimization results based on NPC and COE

Scenario	Hybrid system configuration	NPC (\$)	COE (\$/kWh)
I	PV/BAT/CONV	11,502	0.216
II	PV/DG/BAT/CONV	13,047	0.245
III	PV/WT/BAT/CONV	13,643	0.257
IV	PV/WT/DG/BAT/CONV	15,749	0.296

Table 2: Optimized system component sizing and their cost implication

Scenario	PV (kW)	WT (Units)	DG (kW)	BAT (Units)	Conv (kW)	O&M (\$/Yr)	Capital (\$)
I	6.68	-	-	14	1.70	219,28	8,667
II	7.98	-	1.50	14	1.73	221.06	10,190
III	6.00	1	-	12	1.94	234.59	10,611
IV	9.60	1	1.50	10	1.77	274.13	12,206

In Fig. 7-10, the monthly electric power production of all four hybrid system configurations is shown. The energy production is measured in MWh for the four different scenarios from January to December. In scenario I as shown in Fig. 7, the power supply was entirely produced by the solar PV system backup by the battery storage system while in scenario II presented in Fig. 8, PV and DG system provided the needed electricity but the potential electrical power generated by the DG is just 1.5kW compared to 7.98kW provided by the PV system. In scenario III, a combination of solar PV and wind turbine system generate the required electric power as shown in Fig. 9. The monthly electric power production of the hybrid PV/WT/DG/BAT/CONV shown in Fig. 10 presented that an integrated system of PV, wind turbine and DG combined together for power generation.

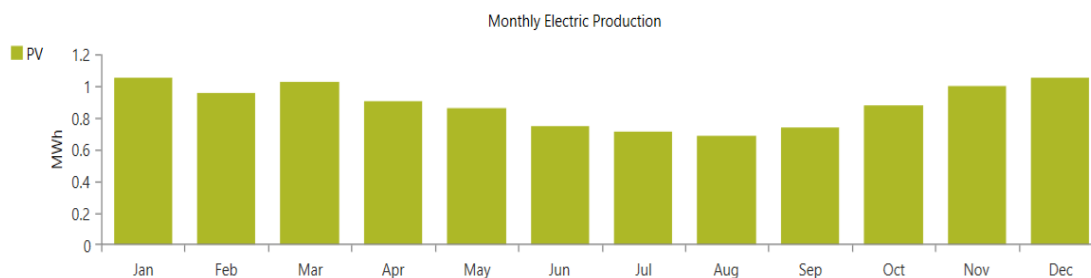


Fig. 7: Monthly electric power production of the hybrid PV/BAT/CONV

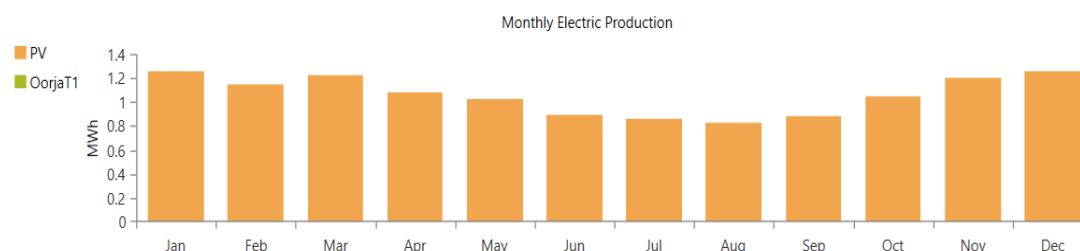


Fig. 8: Monthly electric power production of the hybrid PV/DG/BAT/CONV

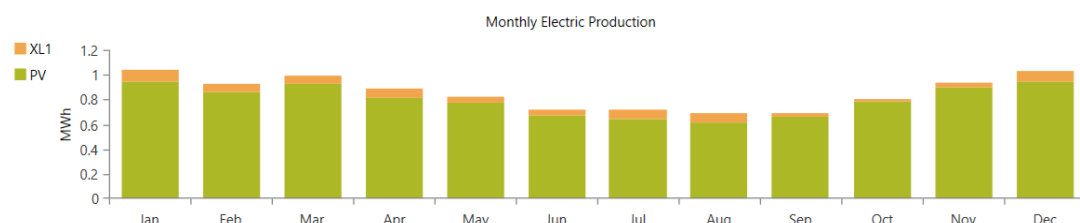


Fig. 9: Monthly electric power production of the hybrid PV/WT/BAT/CONV

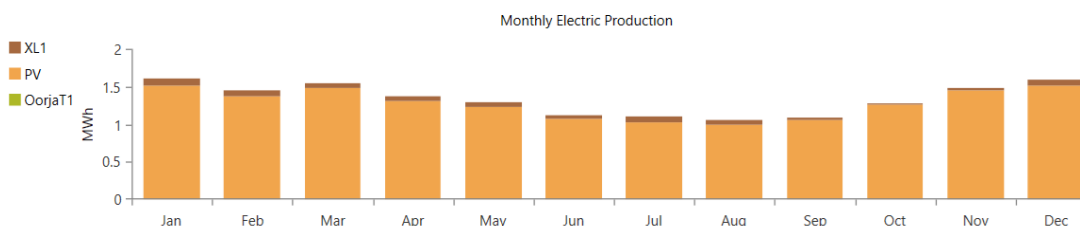


Fig. 10: Monthly electric power production of the hybrid PV/WT/DG/BAT/CONV

Technoeconomic Analysis of Scenario I

In this scenario, the hybrid system configuration comprises solar PV, a battery storage system and converters. The renewable electricity percentage penetration of this system is 100%. However, the NPC and COE are respectively \$11,502 and \$0.216/kWh for the hybrid system configuration. This cost is the lowest among all other optimization results obtained in this study. Also, the system shows an unmet electric load of 0.0756% per year, which is a very negligible value considering the fact that it is far less than 1%. Therefore, this hybrid

integrated technology is essentially the most techno-economically viable system from different standpoints.

Technoeconomic Analysis of Scenario II

In this second scenario, the system configures PV/DG/BAT/CONV with an NPC of \$13,047 and a COE of \$0.245/kWh. It is the second most feasible system from a techno-economic perspective. The solar PV power generating component of the integrated system supplies almost all the electric power required by the load. From the perspective of the COE, there is a percentage difference of approximately 12% higher than the system presented in scenario I. Thus, this system cannot be considered as an optimal option due to the higher value of cost recorded compared to PV/BAT/CONV.

Technoeconomic Analysis of Scenario III

In scenario III, the hybrid combination of renewable energy systems of PV/WT/BAT/CONV was presented. The NPC and the COE obtained by the optimization of this system are quite higher than the system configured in both scenarios I and II. On this account, the hybrid configuration of PV/WT/BAT/CONV cannot be considered optimal due to the recorded NPC of \$13,643 and COE of \$0.257/kWh. In this scenario, the PV still provides the largest share of the electric power generation compared to the integrated wind energy turbine system.

Technoeconomic Analysis of Scenario IV

This configuration of a hybrid system is based on the utilization of PV, wind turbine, diesel generator, battery storage and converters. This hybrid system with an NPC = \$15,749 and \$0.296/kWh, thereby making it the least performing technoeconomic hybrid power system in this study. Compared with the hybrid configuration of the system in scenario I, there is a huge cost gap in the NPC and COE, which are \$4,247 and \$0.08/kWh, respectively. This configuration presented the optimization with the greatest NPC and COE, thereby making it economically worst economic scenario for the electrification of the case study scenario.

Conclusion

This study deals with the optimal technoeconomic configurations of small-scale hybrid energy generators for rural electrification. The system integrated comprises solar PV, WT, DG, BAT and CONV in four different configurations of PV/BAT/CONV, PV/DG/BAT/CONV, PV/WT/BAT/CONV and PV/WT/DG/BAT/CONV. The system is designed in a different configuration, scenario I, which utilizes the integration of PV/BAT/CONV ranks as the best configuration concerning the NPC and COE with their respective values of \$11,502 and \$0.216/kWh based on 100% renewable energy fraction. This indicates that the system has a reasonable environmental sustainability index, considering the fact that no emission is produced. The financial viability of this system over the 25-year lifespan of the project

makes it an economically reliable system among the three other systems. The simulation results indicated that the system has the potential capability to produce sufficient energy to cater to the electric power demand of the community with nearly negligible annual unmet electricity of 0.0756%. Consequently, the system presented in this study can be replicated for the design and optimization of the electrification of any off-grid rural community. In addition, further research can be accomplished for future work to include other sources of renewable energy schemes, such as small-scale hydropower and the application of biomass technology for power generation.

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